

Sold Listings: Exploratory Data Analysis Report

This report will observe the Sold dataset and prepare it for data analysis.

Dataset Overview

```
Rows in Sold Listing Dataset: 639878
```

```
Columns in Sold Listing Dataset: 82
```

```
Index(['BuyerAgentAOR', 'ListAgentAOR', 'Flooring', 'ViewYN', 'WaterfrontYN',  
      'BasementYN', 'PoolPrivateYN', 'OriginalListPrice', 'ListingKey',  
      'ListAgentEmail', 'CloseDate', 'ClosePrice', 'ListAgentFirstName',  
      'ListAgentLastName', 'Latitude', 'Longitude', 'UnparsedAddress',  
      'PropertyType', 'LivingArea', 'ListPrice', 'DaysOnMarket',  
      'ListOfficeName', 'BuyerOfficeName', 'CoListOfficeName',  
      'ListAgentFullName', 'CoListAgentFirstName', 'CoListAgentLastName',  
      'BuyerAgentMlsId', 'BuyerAgentFirstName', 'BuyerAgentLastName',  
      'FireplacesTotal', 'AssociationFeeFrequency', 'AboveGradeFinishedArea',  
      'ListingKeyNumeric', 'MLSAreaMajor', 'TaxAnnualAmount',  
      'CountyOrParish', 'MlsStatus', 'ElementarySchool', 'AttachedGarageYN',  
      'ParkingTotal', 'BuilderName', 'PropertySubType', 'LotSizeAcres',  
      'SubdivisionName', 'BuyerOfficeAOR', 'YearBuilt', 'StreetNumberNumeric',  
      'ListingId', 'BathroomsTotalInteger', 'City', 'TaxYear',  
      'BuildingAreaTotal', 'BedroomsTotal', 'ContractStatusChangeDate',  
      'ElementarySchoolDistrict', 'CoBuyerAgentFirstName',  
      'PurchaseContractDate', 'ListingContractDate', 'BelowGradeFinishedArea',  
      'BusinessType', 'StateOrProvince', 'CoveredSpaces',  
      'MiddleOrJuniorSchool', 'FireplaceYN', 'Stories', 'HighSchool',  
      'Levels', 'LotSizeDimensions', 'LotSizeArea', 'MainLevelBedrooms',  
      'NewConstructionYN', 'GarageSpaces', 'HighSchoolDistrict', 'PostalCode',  
      'AssociationFee', 'LotSizeSquareFeet', 'MiddleOrJuniorSchoolDistrict',  
      'OriginatingSystemName', 'OriginatingSystemSubName',  
      'BuyerAgencyCompensationType', 'BuyerAgencyCompensation'],  
      dtype='object')
```

Important note: While loading the monthly CSV files into a Pandas dataframe, several columns ran into a DtypeWarning.

```
/var/folders/86/cdz_5pvx705dzpfgv_0qsc600000qn/T/ipykernel_70571/1828484726.py:12:  
DtypeWarning: Columns (2,36,39,56,74) have mixed types. Specify dtype option on  
import or set low_memory=False.  
df = pd.read_csv(filename)  
/var/folders/86/cdz_5pvx705dzpfgv_0qsc600000qn/T/ipykernel_70571/1828484726.py:35:  
DtypeWarning: Columns (2,36,39,56,74) have mixed types. Specify dtype option on  
import or set low_memory=False.  
sold = pd.read_csv(sold_filename)  
/var/folders/86/cdz_5pvx705dzpfgv_0qsc600000qn/T/ipykernel_70571/1828484726.py:35:  
DtypeWarning: Columns (4) have mixed types. Specify dtype option on import or set
```

```
low_memory=False.
sold = pd.read_csv(sold_filename)
/var/folders/86/cdz_5pvx705dzpfgv_0qsc600000qn/T/ipykernel_70571/1828484726.py:35:
DtypeWarning: Columns (4,74) have mixed types. Specify dtype option on import or set
low_memory=False.
sold = pd.read_csv(sold_filename)
```

Columns with mixed values are defaulted to object type. However, it is important to understand the expected data types of each column and make adjustments accordingly. Since there are 82 columns, it is efficient to perform this after feature selection (or when each feature is analyzed).

Data Filtering

Observe the following property types:

```
array(['Residential', 'CommercialLease', 'Land', 'ResidentialLease',
      'ManufacturedInPark', 'ResidentialIncome', 'CommercialSale',
      'BusinessOpportunity'], dtype=object)
```

And filter to Residential.

```
Sold rows before filtering: 639878
Sold rows after filtering: 430437
```

Data Cleaning

Validate completeness

Out of the 82 columns, the following are missing more than 90% of values:

	Missing Count	Missing Percent
WaterfrontYN	430166	99.94
BasementYN	422000	98.04
FireplacesTotal	430437	100.00
AboveGradeFinishedArea	430437	100.00
TaxAnnualAmount	430437	100.00
BuilderName	409366	95.10
TaxYear	430437	100.00
BuildingAreaTotal	400265	92.99
ElementarySchoolDistrict	430437	100.00
CoBuyerAgentFirstName	391163	90.88
BelowGradeFinishedArea	427911	99.41
BusinessType	430437	100.00
CoveredSpaces	430437	100.00
LotSizeDimensions	409538	95.14
MiddleOrJuniorSchoolDistrict	430437	100.00

Before concatenating the monthly datasets, two features were intentionally removed. Latfilled and Lonfilled were only recorded by 7 datasets, which guarantees that the final dataset will be missing these two columns for more than 90% of the rows.

Based on the above table, all but BuildingAreaTotal can also be dropped. Many of these features are categorized as metadata, meaning they don't hold meaningful contributions to the property value overall. Furthermore, columns with 100% missing values don't have any contribution to analysis. For features like WaterfrontYN and BasementYN, these are dropped because they are considered insufficiently complete for reliable exploratory analysis. BuildingAreaTotal can be retained as it is considered a market analysis feature and has the potential to provide meaningful contributions to analysis.

* LotSizeDimensions is dropped because the dataset already records LotSizeAcre and LotSizeSquareFeet.

Key Numeric Fields

Analyze the distribution of key numeric fields: ClosePrice, ListPrice, OriginalListPrice, LivingArea, LotSizeAcres, BedroomsTotal, BathroomsTotalInteger, DaysOnMarket, and YearBuilt.

For each field, generate histograms, boxplots, and percentile summaries, and identify extreme outliers for later handling. Ensure that the features are their expected data type prior.

```
ClosePrice          float64
ListPrice           float64
OriginalListPrice   float64
LivingArea          float64
LotSizeAcres        float64
BedroomsTotal       float64
BathroomsTotalInteger float64
DaysOnMarket        int64
YearBuilt           float64
dtype: object
```

The features do not require changes to data type.

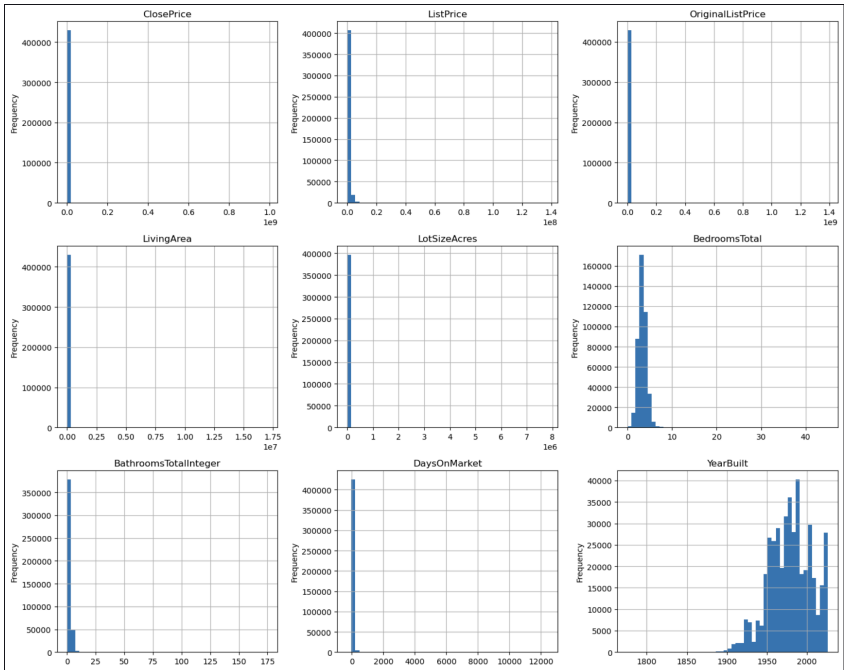
Percentile Summaries

	ClosePrice	ListPrice	OriginalListPrice	LivingArea	LotSizeAcres	BedroomsTotal	BathroomsTotalInteger	DaysOnMarket	YearBuilt
count	4.304350e+05	4.304370e+05	4.296470e+05	4.301920e+05	3.967880e+05	430425.000000	430367.000000	430437.000000	430055.000000
mean	1.193116e+06	1.141732e+06	1.226553e+06	1.904057e+03	6.400482e+01	3.203239	2.536668	37.333535	1978.600830
std	6.174142e+06	1.357896e+06	6.686765e+06	2.596825e+04	1.568755e+04	1.066820	1.132613	53.668295	26.292955
min	0.000000e+00	5.250000e+02	0.000000e+00	0.000000e+00	0.000000e+00	0.000000	0.000000	-288.000000	1776.000000
25%	5.750000e+05	5.780000e+05	5.850000e+05	1.248000e+03	1.200000e-01	3.000000	2.000000	8.000000	1960.000000
50%	8.250000e+05	8.190000e+05	8.280000e+05	1.644000e+03	1.665000e-01	3.000000	2.000000	18.000000	1979.000000
75%	1.300000e+06	1.295000e+06	1.299000e+06	2.221000e+03	2.732000e-01	4.000000	3.000000	48.000000	1999.000000
max	9.895000e+08	1.375000e+08	1.390000e+09	1.702132e+07	7.810698e+06	45.000000	175.000000	12430.000000	2026.000000

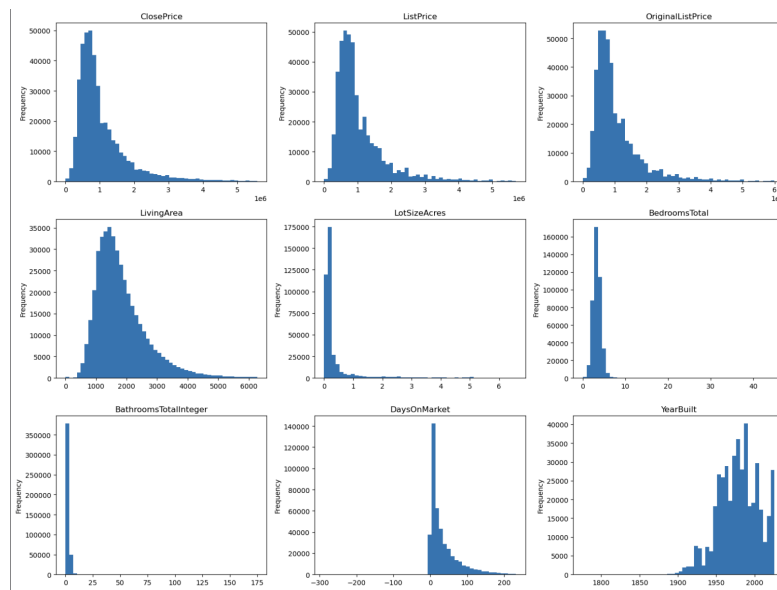
Extreme Outliers

	Feature	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count	Outlier Percent
0	ClosePrice	575000.00	1300000.00	725000.00	-512500.00	2387500.0	32116	7.46
1	ListPrice	578000.00	1295000.00	717000.00	-497500.00	2370500.0	32426	7.53
2	OriginalListPrice	585000.00	1299000.00	714000.00	-486000.00	2370000.0	33657	7.82
3	LivingArea	1248.00	2221.00	973.00	-211.50	3680.5	18903	4.39
4	LotSizeAcres	0.12	0.27	0.15	-0.11	0.5	61733	14.34
5	BedroomsTotal	3.00	4.00	1.00	1.50	5.5	23580	5.48
6	BathroomsTotalInteger	2.00	3.00	1.00	0.50	4.5	19879	4.62
7	DaysOnMarket	8.00	48.00	40.00	-52.00	108.0	32989	7.66
8	YearBuilt	1960.00	1999.00	39.00	1901.50	2057.5	1015	0.24

Histograms



Removing the extreme outliers from plots:



Boxplots

